

Sean Girgis

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Senior Data Engineer | Capacity Planning Performance and AI-Driven Infrastructure Optimization

Executive Profile

Senior data, capacity, and performance engineer with 20+ years delivering reliable data pipelines, forecasting, observability, and infrastructure analysis in regulated and high-scale enterprises. Combines Python, SQL, ETL, cloud data services, and application performance management with practical AI automation to improve operational decisions and reduce manual work.

Core Competencies

Data Engineering and ETL | Capacity Planning and Forecasting | Performance Engineering | Python and SQL | AWS and PySpark | Data Quality | Observability and APM | AI-Assisted Automation | Technical Leadership

Professional Experience

LTIMindtree | Capacity and Performance Specialist

2026 - Present

- Apply AI-assisted automation to capacity reporting and recurring operational workflows, integrating data from JIRA, Microsoft SQL Server, and Splunk to improve repeatability and report quality.
- Use Copilot, AI, and Python to organize, catalog, and support migration of a large Power BI report estate, making reports easier to research and manage.
- Research an LLM-enabled team knowledge-base and analyst command-center concept supported by retrieval, Python, and PowerShell; the initiative is investigative and not represented as production.

Citi | Senior Capacity and Data Engineer

Nov 2017 - Dec 2025

- Architected automated Python and Pandas ETL pipelines for P95 telemetry from 6,000+ endpoints, replacing manual processes and strengthening validation, cleansing, and anomaly detection.
- Designed Oracle schemas for historical retention and unified reporting, enabling seasonal analysis, executive dashboards, and capacity-risk forecasting.
- Developed Prophet and scikit-learn models to predict bottlenecks three to six months ahead and inform provisioning, consolidation, and rightsizing decisions.
- Managed enterprise capacity requests, monthly reporting, peak-period analysis, and observability across BMC TrueSight, CA APM, and AppDynamics.

G6 Hospitality | Performance Engineer

Mar 2017 - Nov 2017

- Managed Dynatrace AppMon and Synthetics for critical digital systems; led performance-data mining, bottleneck analysis, an upgrade from 6.5 to 7.0, TLS 1.2 enablement, and AWS migration support.

HCL / Entergy | APM Consultant

Jan 2017 - Mar 2017

- Supported CA Application Performance Management, Customer Experience Manager, and Application Delivery Analysis for utility systems.

CA Technologies / TIAA-CREF | Senior Consultant and CA APM Subject Matter Expert

2011 - 2016

- Led enterprise CA APM implementations and upgrades across environments of roughly 4,000-6,000 agents and 50+ Enterprise Managers.
- Designed monitoring modules, dashboards, alerts, Golden Images, and Perl/KornShell extraction scripts; provided sizing guidance, troubleshooting, documentation, and client training.

AT&T | Performance Test Engineer**Aug 2010 - Jul 2011**

- Analyzed J2EE telecommunications applications under load, documented JDBC, thread, memory, CPU, and garbage-collection behavior, and automated monitoring with JMX and Wily Introscope.

Sabre | Senior Systems and Data Migration Engineer**May 2008 - 2010**

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a six-node Oracle RAC cluster; optimized C++/OCI processing and reduced hardware footprint by approximately 95% while maintaining sub-second latency.

Earlier Experience | Software Architecture Development and Support**1996 - 2008**

- Built and supported high-availability C++ systems, billing interfaces, database and batch processes, and Linux/Unix operations across government, telecom, and enterprise environments.

Selected Technical Projects

- AI Job Search Pipeline (2026): Built a Python agentic workflow with FAISS semantic matching, LLM scoring and tailoring, document generation, quality gates, MCP integrations, and tracking across 100+ postings.
- HorizonScale (2025): Built a Python/PySpark forecasting project with Prophet, scikit-learn, and Streamlit; reduced forecasting cycles by approximately 90% and projected bottlenecks up to six months ahead.
- Serverless Lakehouse: Designed an AWS S3, Glue, Athena, and Bedrock platform with a text-to-SQL agent and Parquet/Snappy storage optimization.

Technical Skills

Data: Python, SQL, Oracle, PL/SQL, Pandas, ETL/ELT, PySpark/Spark, Airflow, Parquet, dimensional modeling, data quality

Cloud and platforms: AWS S3, Glue, Athena, Bedrock; Docker; Linux/Unix; Streamlit

Capacity and observability: BMC TrueSight/TSCO, CA APM/Introscope, AppDynamics, Dynatrace, Splunk, Prophet, scikit-learn

AI and automation: LLM agents, RAG, FAISS, MCP, Copilot, AI-assisted development, PowerShell

Education and Certifications

Post-Graduate Diploma, Computer Engineering Technology / Computer Science - Humber College, Toronto | Bachelor of Science, Civil Engineering - Zagazig University, Egypt | Brainbench certifications in C++, C, Java, Object-Oriented Design, and Mathematics